



CS 4445 – Data Communication and Networking

Prerequisites: CS 3325 or CS 3330.

Course description

This course provides a comprehensive introduction to the principles, architecture, and operation of modern computer networks. Using a top-down approach, it begins with real-world network applications and progressively explores the underlying technologies that make data communication possible—from the application layer down to the physical layer.

Students will learn how information is transmitted across the Internet, how protocols such as HTTP, DNS, TCP/IP, and Ethernet function, and how networks are designed, managed, and secured. The course covers both theoretical concepts and practical skills, including socket programming, packet analysis (Wireshark), and network troubleshooting.

Emphasis is placed on understanding the layered structure of network protocols, data transfer reliability, routing and addressing, congestion control, wireless communication, and network security. Emerging topics such as software-defined networking (SDN), 5G, and Internet of Things (IoT) are also introduced to give students insight into the future of networking technologies.

By the end of the course, students will have both a conceptual understanding and hands-on experience in analyzing, designing, and implementing computer network systems.

Course Objectives

By the end of this course, students should be able to:

1. Understand the layered architecture of computer networks and how various protocols interact.
2. Analyze and explain key protocols at the Application, Transport, Network, Link, and Physical layers.
3. Design or reason about network applications (e.g. HTTP, DNS, P2P) and implement basic client-server programs.
4. Explain reliable data transfer, congestion control, and flow control mechanisms in TCP/UDP.
5. Understand routing algorithms, forwarding, addressing, NAT, and IP (IPv4/IPv6).
6. Understand link-layer protocols, multiple access, Ethernet, ARP, switches, VLANs, and error detection/correction.
7. Explore wireless and mobile networking.

8. Understand network security principles (cryptography, authentication, TLS, firewalls) in modern networks.
9. (Optional / advanced) Learn about network control, software-defined networking (SDN), network management, and the softwareization of network functions

Learning Outcomes

- You will be able to describe the functionality and design of major network protocols (HTTP, TCP, IP, Ethernet, WiFi).
- You can reason about performance — latency, throughput, congestion, loss tradeoffs.
- You will have hands-on experience in network programming or packet-level experimentation.
- You can analyze network topologies, routing decisions, address assignment, NAT, and switching behavior.
- You will understand security mechanisms and their use in real-world protocols.
- For advanced students, you gain insight into SDN, network control, and future networking trends.

Textbook:

Computer Networking – A top-down approach, 9th Edition, Pearson. Jame F. Kurose, Keith W. Ross.

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Grading Scale

90-100 - A
80- 89 - B
70- 79 - C
60- 69 - D
Below 60 - F

Grading Computation:

Class discussion: 15%
Homeworks: 10% (2x5%)
Group project: 15%
Midterm test 25% (2-hour)
Final Exam: 35% (2-hour)

ATTENDANCE

Student's attendance will be marked for every lesson. A student who is absent more than **03 times** with or without faculty's permission will get F for the course.

SUBMITTING ASSIGNMENTS AND PROJECTS:

The homeworks and project must be completed using **Microsoft WORD 20xx, Times New Roman, 12-14pt. font, with standard margins, and 5-space paragraph indentions.** The homework files with format: *Std_name_Std_ID_Homework#X*. Points will be deducted for failure to follow the format requirements and submission procedures.

The **deadline** for students to hand in their home work is by **5:00 p.m** of the day assigned in the time table.

LATE WORK POLICY:

Late work is not accepted unless **extraordinary circumstances** exist, such as student hospitalization.

If I have not heard from you regarding your “**extraordinary circumstances**” by the deadline for assignments, test projects, or quizzes, no make-up work will be allowed. Requests for extensions must be made **in advance** and accompanied by appropriate written documentation if the excuse is acceptable to the instructor. **"Computer problems" are not an acceptable excuse.**

Missing any part of the course agenda may prevent completion of the course. If you know of any difficulty of any type (i.e., an illness, employment change, TDY, etc.) which may prevent completion of this course, notify the instructor as soon as possible. Failure to do so will result in failure for an assignment and/or failure of the course.

EXAMINATION SCHEDULE & INSTRUCTIONS:

The Final Exam will consist of a two-hour, objective Exam based on the material learned in the course.

ACADEMIC HONESTY AND PLAGIARISM:

In accordance with the **Standards of Conduct** in the Troy University Bulletin, a student or organization may be disciplined, up to and including expulsion, if deemed in violation of the **Standards of Conduct** for the commission of or the attempt to commit this offense: Dishonesty, such as cheating, plagiarism or knowingly furnishing false information to the university, faculty, or other officers or employees of the university.

The awarding of a university degree attests that an individual has demonstrated mastery of a significant body of knowledge and skills of substantive value to society. Any type of dishonesty in securing those credentials therefore invites serious sanctions, up to and including suspension and expulsion (see Standard of Conduct in each TROY *Catalog*). Examples of dishonesty include actual or attempted cheating, plagiarism*, or knowingly furnishing false information to any university employee.

*Plagiarism is defined as submitting anything for credit in one course that has already been submitted for credit in another course, or copying any part of someone else's intellectual work – their ideas and/or words – published or unpublished, including that of other students, and portraying it as one's own. Proper quoting, using strict APA formatting, is required, as described by the instructor. All students are required to read the material presented at: <http://troy.troy.edu/writingcenter/research.html>

- Students must properly cite any quoted material. No term paper, business plan, term project, case analysis, or assignment may have **no more than 20% of its content**

quoted from another source. Students who need assistance in learning to paraphrase should ask the instructor for guidance and consult the links at the Troy Writing Center.

- This university employs **plagiarism-detection software**, through which *all* written student assignments are processed for comparison with material published in traditional sources (books, journals, magazines), on the internet (to include essays for sale), and papers turned in by students in the same *and* other classes in this *and all previous terms*. The penalty for plagiarism may range from zero credit on the assignment, to zero in the course, to expulsion from the university with appropriate notation in the student's permanent file.

Plagiarism Policy: Plagiarism is the passing off of the thoughts or works of another as one's own. Plagiarism involves giving the impression that a person has thought, written or produced something that has, in fact, been borrowed from another. Plagiarism may result from poor technique or more serious causes such as: copying the work of another person; submitting the work of another person; or closely paraphrasing a piece of work without due acknowledgment.

Allegations of Plagiarism: Depending on the circumstances, the penalty imposed may include a warning, resubmission, loss of marks, a failing grade on a particular assignment or course, or a charge of misconduct to be dealt with by Troy University.

Artificial Intelligence (AI) Policy: In this course, we will explore how to use AI tools to enhance our learning. You are encouraged to use AI tools such as GitHub Copilot, Grammarly, ChatGPT, or others that support your understanding of the material.

For assignments, I ask that you first attempt the questions on your own. If you're unsure how to begin, you may use AI to request a single hint and try again. If that doesn't help, you may continue asking for hints until you're able to solve the problem. Important: If you use AI for any part of an assignment, you must clearly indicate which parts were assisted by AI and which tool you used. You will not lose any points for getting help from AI—as long as you are transparent about it.

TECHNOLOGY REQUIREMENTS:

Students must have:

- Access to the Internet;
- E-mail software capable of sending and receiving attached files;
- Microsoft WORD 201X software;
- Network simulation, packet analysis (Wireshark), Python;
- Virus protection software, installed and active, to prevent the spread of viruses via the Internet and email. It should be continually updated!

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Week	Chapter	Topic
1. 09/10/2026	1	Computer networks and the Internet
2. 16/10	2	Appliacation layer
3. 23/10	3	Application layer (cont) Transport layer
4. 30/10	3	Transport layer (cont)
5. 6/11	4	The network layer: Data plane Group Project handout Homework #1
6. 13/11	5	The network layer: Control plane
7. 20/11	1,2,3,4,5	Discuss chapter 1, 2, 3, 4, 5. Midterm test
8. 27/11	6	The link layer and LANs
9. 04/12	7	Wireless and mobile networks
10. 11/12	8	Wireless and mobile networks (cont) Homework #2
11. 18/12	8	Security in computer networks
12. 25/12		Security in computer networks (Cont)
13. 01/01/2026	6,7,8,9	Discuss chapter 6,7,8, 9 Revision, Q&A
14. 08/01/2026		Final Exam
15. 15/01/2026		Project Presentation